

Task 1 – Installation of Hypervisors and Initiation of VMs

Objective

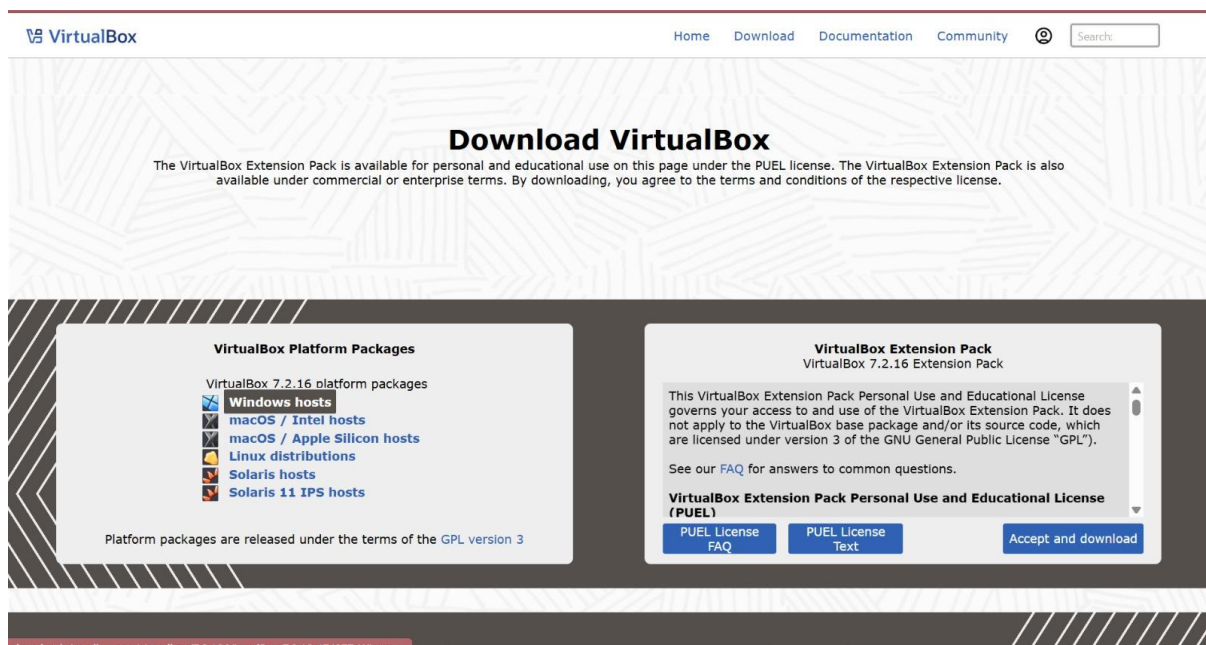
To install and configure an open-source hypervisor, create and configure a Virtual Machine (VM) using an existing Ubuntu ISO image, boot the VM, and verify the Linux environment.

1. Install Oracle VirtualBox

Oracle VirtualBox was installed as the open-source hypervisor.

Command to verify VirtualBox installation

Screenshot :

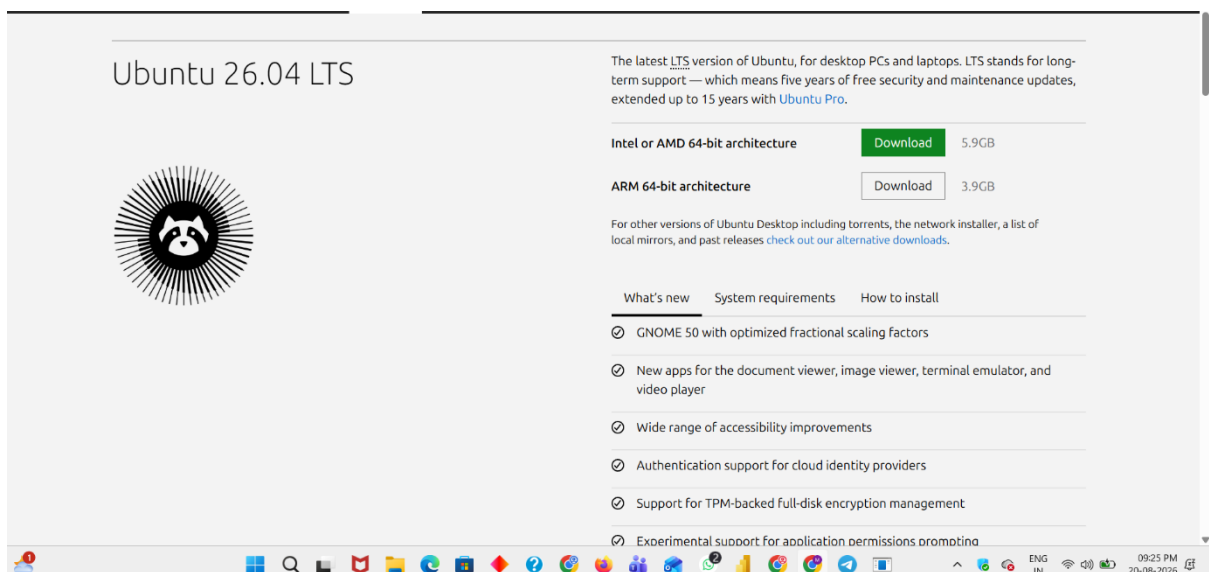


Task 2. Obtain a suitable VM image/ISO file.

Ubuntu 26.04 LTS was selected for the virtual machine.

For an Intel/AMD 64-bit system, the **Intel or AMD 64-bit architecture** ISO was selected.

Screenshot:



Task 3 — Create a New Virtual Machine

Objective

Create a new virtual machine using Oracle VirtualBox.

Steps

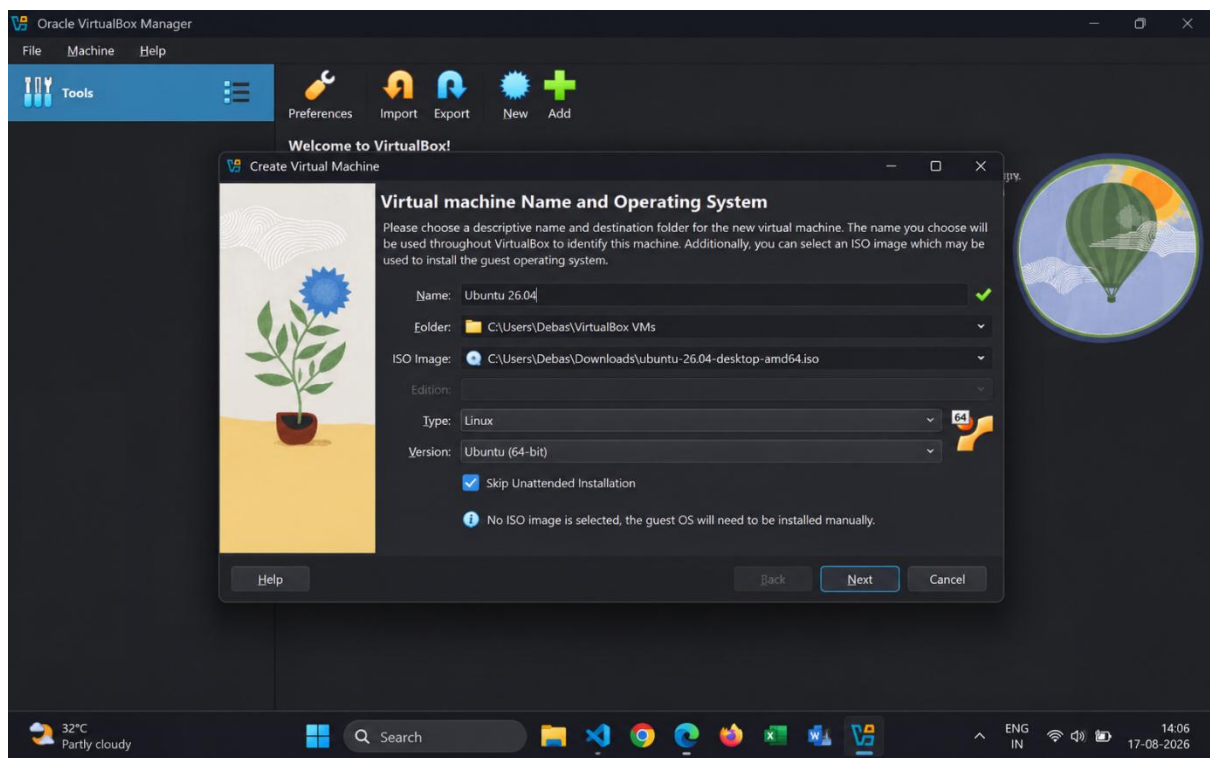
1. Opened Oracle VirtualBox.
2. Clicked **New** to create a new virtual machine.
3. Entered the VM name as Ubuntu 26.04.

4. Selected:

- OS: Linux
- Distribution: Ubuntu
- Version: Ubuntu (64-bit)

5. Proceeded with the VM creation.

Screenshot :



Task 4 — Configure the VM's CPU, RAM, Storage, and Network

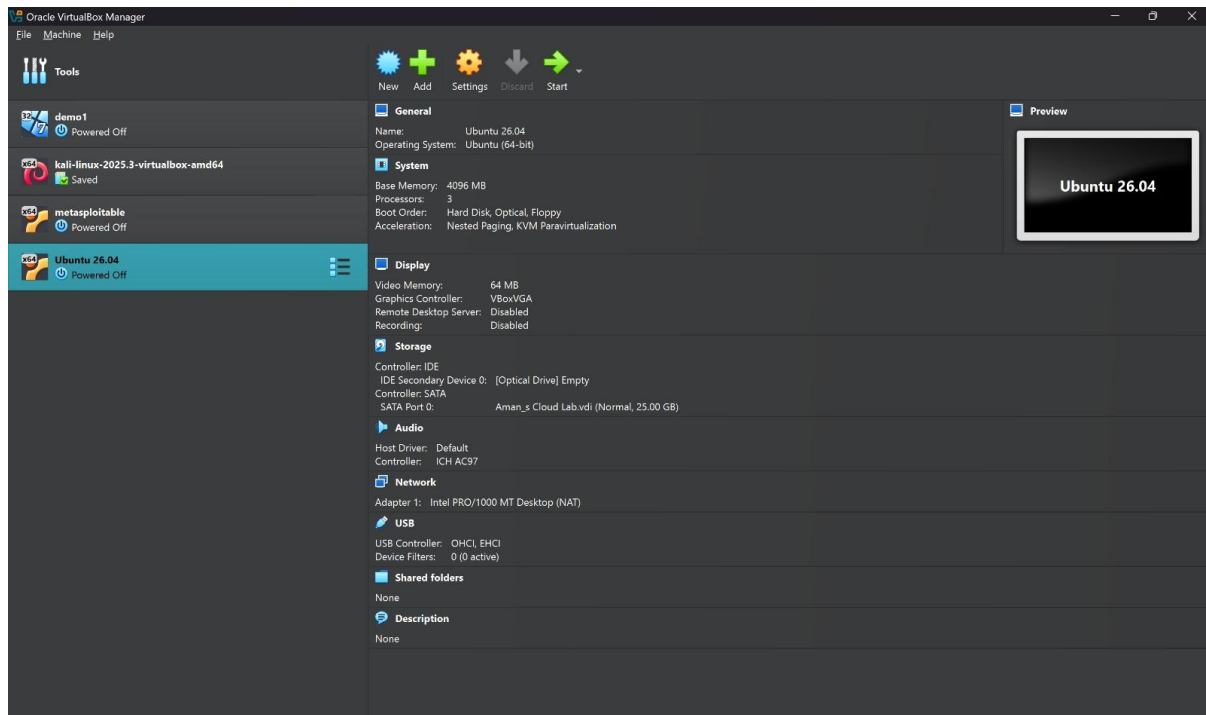
Objective

Configure the virtual machine's hardware resources and network settings using Oracle VirtualBox.

Steps

1. Opened **Oracle VirtualBox Manager**.
2. Selected the **Ubuntu 26.04** virtual machine.
3. Verified the CPU configuration:
 - Processors: **3**
4. Verified the RAM allocation:
 - Base Memory: **4096 MB (4 GB)**
5. Verified the storage configuration:
 - Virtual Hard Disk: **25.00 GB**
 - Controller: **SATA**
6. Verified the network configuration:
 - Adapter: **Intel PRO/1000 MT Desktop**
 - Network Mode: **NAT**
7. Confirmed all the configured settings from the VM's summary panel.

Screenshot:



Task 5 — Attach the Image/ISO File to the VM

Objective

Attach the Ubuntu ISO image to the virtual machine using Oracle VirtualBox.

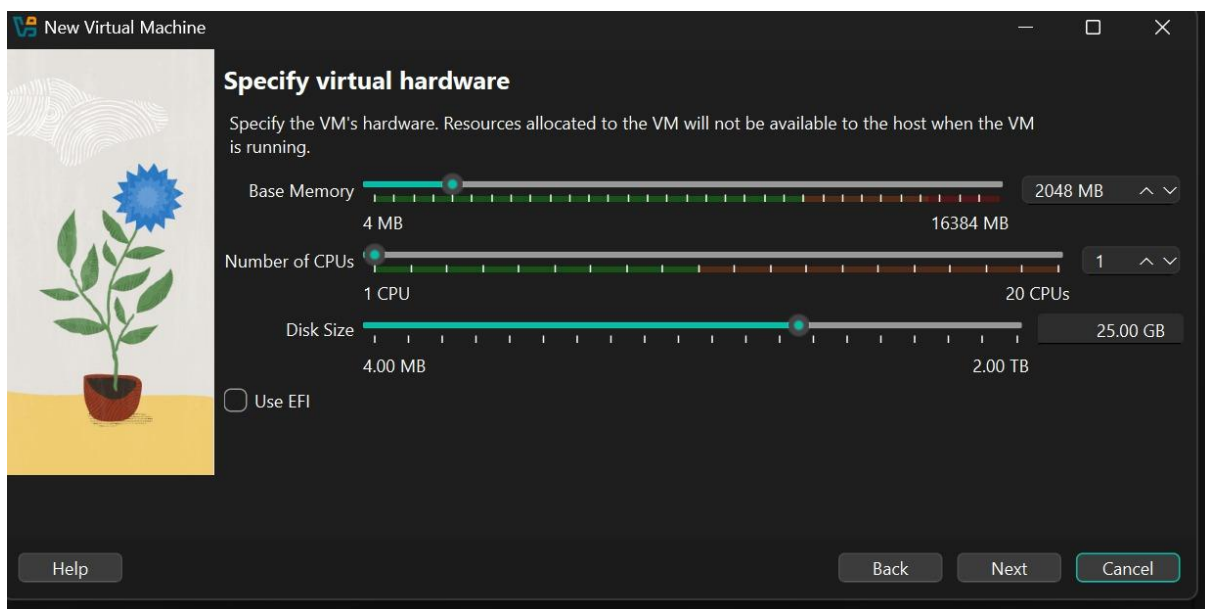
Steps

1. Opened **Oracle VirtualBox Manager**.
2. Selected the **Ubuntu 26.04** virtual machine.
3. Opened the **New Virtual Machine** configuration window.
4. Selected the required **Ubuntu ISO image** in the **ISO Image** field.
5. Verified that the ISO file ubuntu-26.04-desktop-amd64.iso was detected successfully.

6. Confirmed the operating system details:

- OS: **Linux**
- Distribution: **Ubuntu**
- Version: **Ubuntu (64-bit)**

Screenshot:



Task 6 — Start and Boot the Virtual Machine

Objective

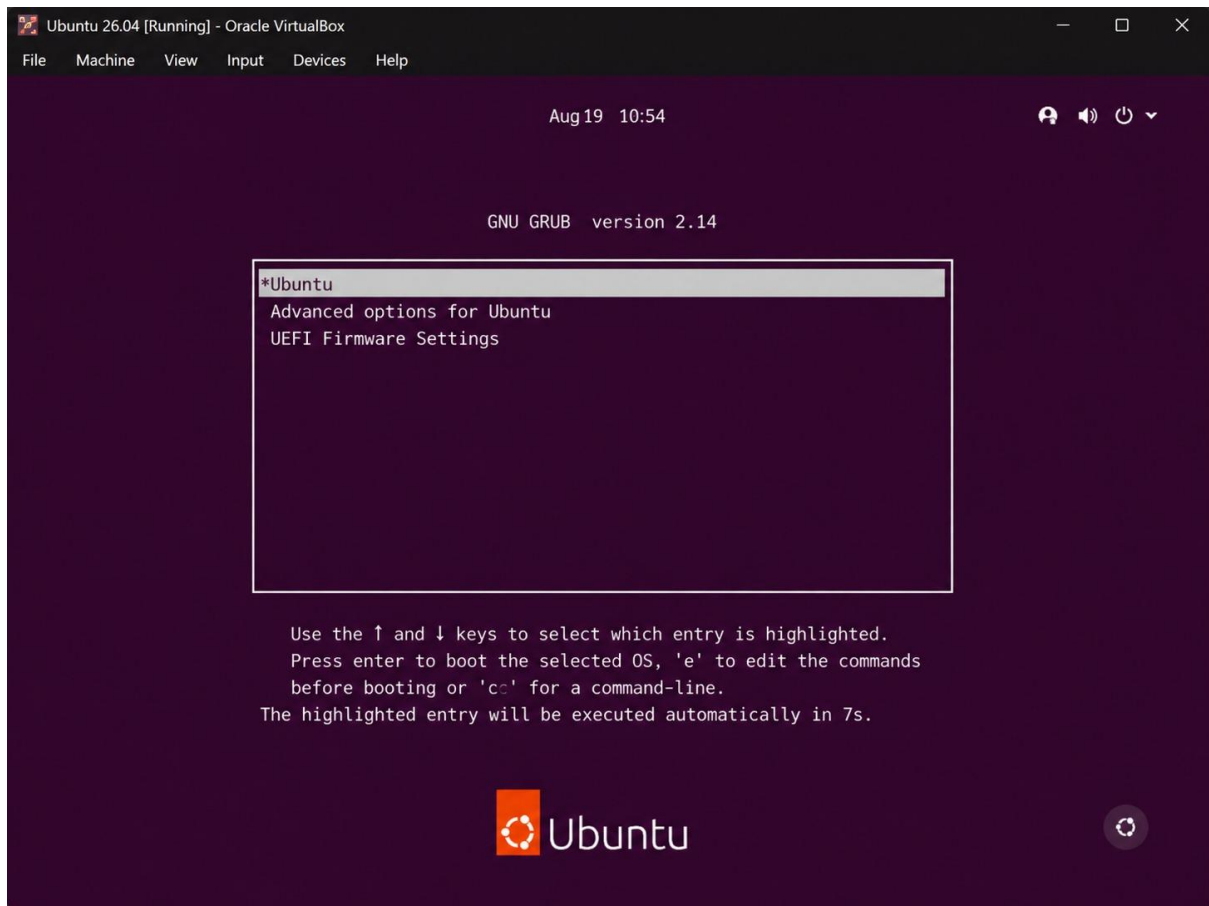
Start the Ubuntu virtual machine and boot the guest operating system successfully.

Steps

1. Opened **Oracle VirtualBox Manager**.

2. Selected the **Ubuntu 26.04** virtual machine.
3. Started the virtual machine.
4. The VM reached the **GNU GRUB boot menu**.
5. The Ubuntu boot options were displayed.
6. Selected the appropriate Ubuntu boot option to continue the boot process.

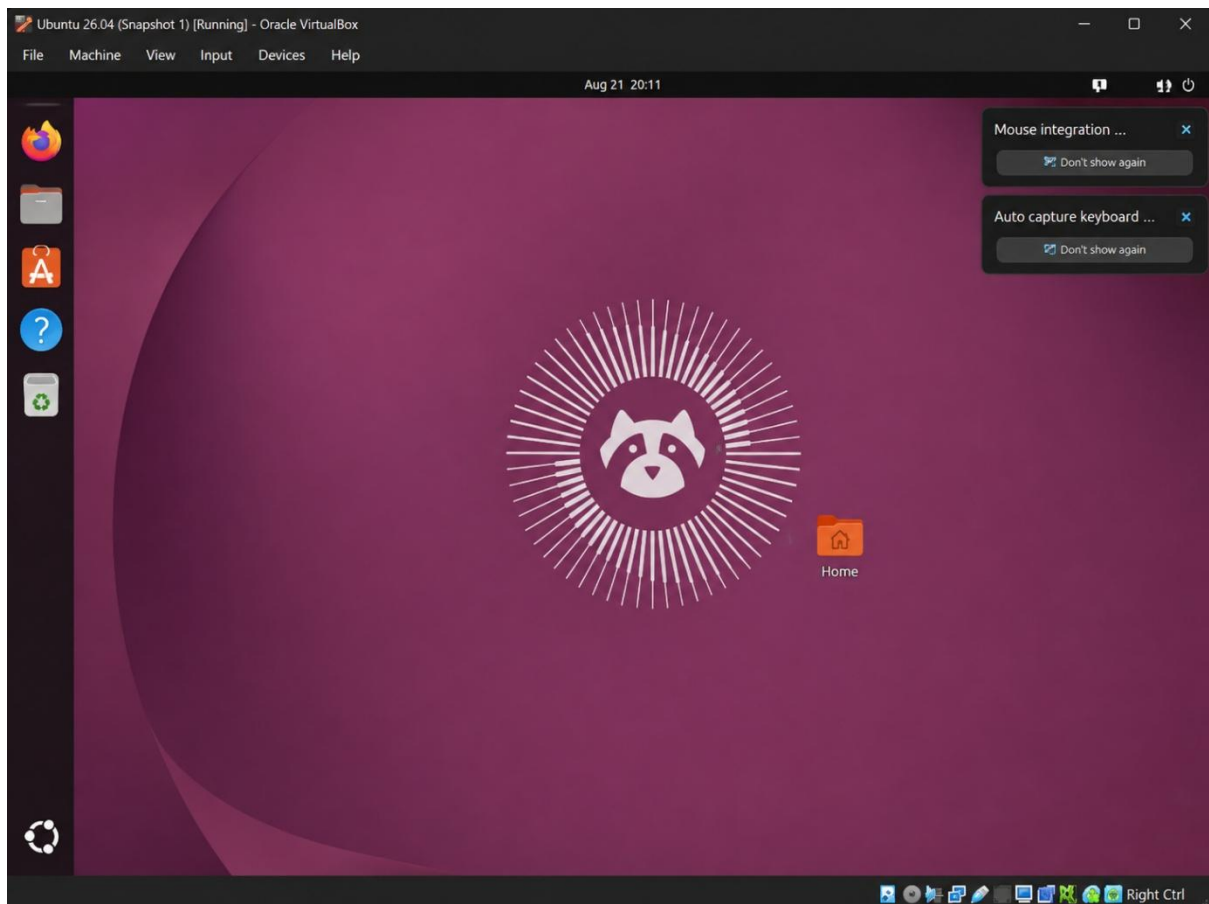
Screenshot :



Task 7 — Complete the Guest OS Installation

- Ubuntu desktop fully loaded
- Left-side Ubuntu application dock visible
- Top bar visible
- Ubuntu wallpaper
- No installation/setup dialog
- VM is running normally inside VirtualBox

Screenshot:



Task 8 — Verify VM and Install Required Build Tools

Objective

Verify that the Ubuntu virtual machine is running and execute the required commands to update the package repositories and install the necessary build tools and Linux kernel headers.

Steps

1. Started the **Ubuntu 26.04** virtual machine in Oracle VirtualBox.
2. Opened the **Terminal** inside the Ubuntu VM.

3. Recorded the terminal output and captured the relevant screenshots for the assignment.
4. The screenshots document the package-management process and the repository access errors encountered during installation.

Screenshots : (next page)

```
Aug 20 14:47
debasmita-bose@debasmita-bose-VirtualBox: ~/Desktop

Errr1: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-15/libgcc-15-dev_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 104.228.246.80]
Errr2: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-15/gcc-15-x86-64-linux-gnu_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr3: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-15/gcc-15_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr4: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-defaults/gcc-x86-64-linux-gnu_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr5: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-defaults/gcc-15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr6: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-15/libstdc++2bxt2b-15-dev_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr7: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-15/gx2bx2b-15-x86-64-linux-gnu_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr8: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-15/gx2bx2b-15_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr9: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-defaults/gx2bx2b-x86-64-linux-gnu_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr10: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/g/gcc-defaults/gx2bx2b_15.2.0-16ubuntu1_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr11: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/m/make-dfsg/make_4.4.1-3_and64.deb 403 Forbidden [IP: 172.66.152.176 80]
Errr12: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/d/dpkg/libdpkg-perl_1.23.7ubuntu1_all.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr13: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/b/bzip2/bzip2_1.0.8-6ubuntu2_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr14: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/l/lto-disabled-list/lto-disabled-list_79_all.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr15: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/d/dpkg/dpkg-dev_1.23.7ubuntu1_all.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr16: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/libx/libxcrypt/libxcrypt-dev_4.5.1-1_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr17: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/b/build-essential/build-essential_12.12ubuntu2.26.04.2_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr18: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/f/fakeroot/libfakeroot_1.37.2-1_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr19: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/f/fakeroot/fakeroot_1.37.2-1_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr20: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/liba/libalgorithm-diff-perl/libalgorithm-diff-perl_1.201-1_all.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr21: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/liba/libalgorithm-diff-xs-perl/libalgorithm-diff-xs-perl_0.04-9_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr22: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/liba/libalgorithm-merge-perl/libalgorithm-merge-perl_0.08-5_all.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr23: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/libf/libfile-fcntllock-perl/libfile-fcntllock-perl_0.22-4ubuntu6_and64.deb 403 Forbidden [IP: 104.20.28.246 80]
Errr24: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/libf/libfile-fcntllock-perl/libfile-fcntllock-perl_0.22-4ubuntu6_and64.deb 403 Forbidden [IP: 104.20.28.246 80]

E: Unable to fetch some archives, maybe run apt update or try with --fix-missing?
debasmita-bose@debasmita-bose-VirtualBox: ~/Desktop$
```

```
Aug 20 14:46
debasmita-bose@debasmita-bose-VirtualBox: ~/Desktop

debasmita-bose@debasmita-bose-VirtualBox:~/Desktop$ sudo apt update
[sudo] password for debasmita-bose:
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Err:2 http://archive.ubuntu.com/ubuntu resolute InRelease
403 Forbidden [IP: 172.66.152.176 80]
Err:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
403 Forbidden [IP: 172.66.152.176 80]
Err:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
403 Forbidden [IP: 172.66.152.176 80]
Err:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Temporary failure resolving 'security.ubuntu.com'
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Err: The repository 'http://archive.ubuntu.com/ubuntu resolute InRelease' is no longer signed.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Err: The repository 'http://archive.ubuntu.com/ubuntu resolute-updates InRelease' is no longer signed.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Err: The repository 'http://archive.ubuntu.com/ubuntu resolute-backports InRelease' is no longer signed.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Err: The repository 'http://security.ubuntu.com/ubuntu resolute-security InRelease' is no longer signed.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
debasmita-bose@debasmita-bose-VirtualBox:~/Desktop$
```

Task 9: Record the VM Configuration

The configuration of the **Ubuntu 26.04 virtual machine** was recorded using Oracle VirtualBox Manager. The VM is configured with **Ubuntu Linux (64-bit)** as the guest operating system and **4096 MB (4 GB)** of base memory. The system configuration also shows the assigned processor and boot device settings.

The storage configuration confirms that the virtual machine is using the **Ubuntu 26.04 virtual hard disk (Ubuntu 26.04.vdi)**, connected through the **SATA controller**. These settings verify the hardware and storage configuration of the created virtual machine.

Figure 9.1: General and System configuration of the Ubuntu 26.04 virtual machine.

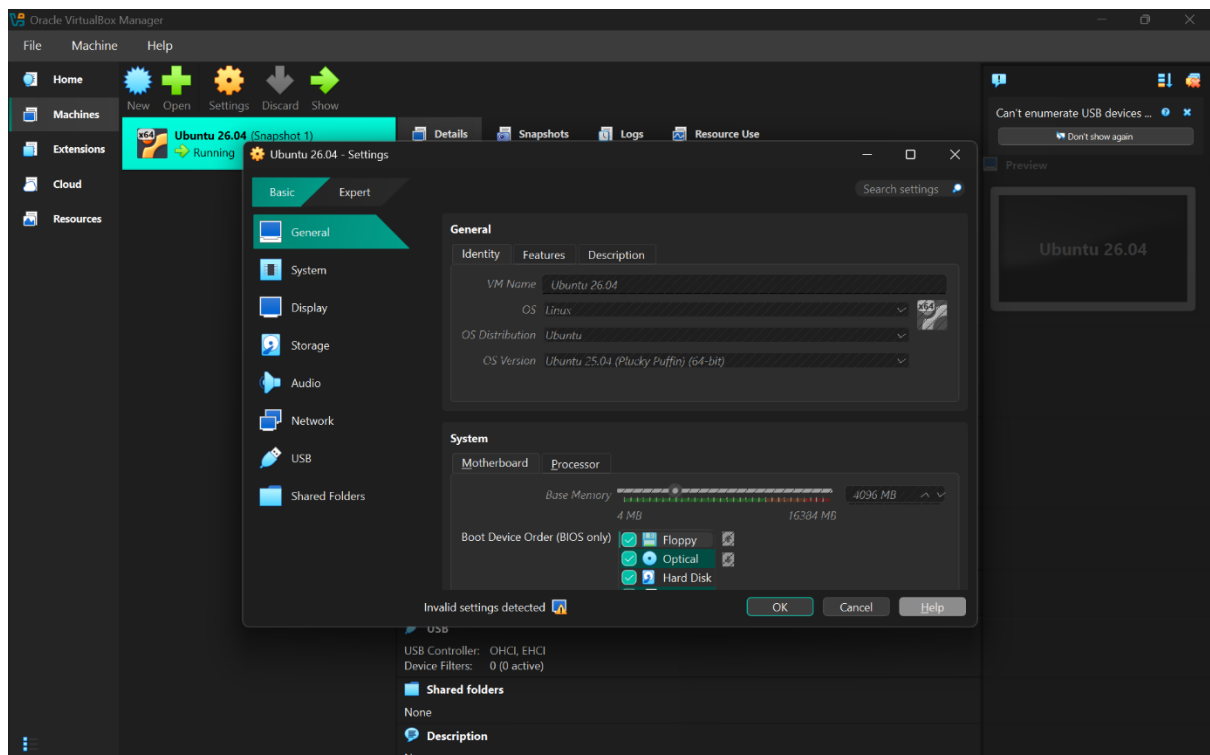


Figure 9.2: Storage configuration showing the attached Ubuntu 26.04.vdi virtual hard disk.

